ActInf MathStream 014.1 ~ Jesse van Oostrum: A concise mathematical de

MathStream | Jesse van Oostrum: A concise mathematical description of active inference | 1:31:14

Hello, welcome everyone. It is May 30th, 2025. We're in Active Inference Math Stream 14.1.
And here with Jesse Van Ustrom discussing a concise mathematical description of active
inference in discrete time. This will be a very useful and epistemic discussion. So, Jesse,
thank you for writing this paper and coming on to you to introduce it and start walking us through it.
Yeah, thank you, Daniel. Thank you for the opportunity of being here and allowing me to
present my paper. Yeah, first of all, I wrote this paper together with my colleague,
Carlotta Langer and my supervisor, Nia Tai. We are all in Hamburg at the Institute of Data Science
Foundations. And yeah, about three years ago, I started getting interested in active inference and
wanted to know really exactly how it works, so to say. But this, I had quite a hard time figuring this
out and that's a bit why this paper came about. And maybe I can, I also have some notes here. I can,
so maybe I can make very specific my question. So what I wanted to know was, okay, we have an agent who has
performed a 1 up to a 1 up to a 1 up to a 1 up to a 1. And then what I wanted to know was an
an agent acting according to active inference. What will its action 80? How will it choose that?
And this question for me was so with the literature at the time, I found it hard to really pin it down.
So to say, so to say, so that was the first main objective of making that more clear and especially
also making it clear in a way that is with in notation that is very accessible to people who have
studied probability theory or any other type of standard mathematics. And yeah. And then throughout the
process, a second question came up. So that was the following. So, so I will go into more detail on this
later, but agent has something that's called a generative model and how to understand that is as
follows. So, uh, the world, we can first look at the real world and that is, uh, generating observations.
For example, observation one world is in world state one, then, uh, observation two. So this is all
happening in the world, but then an agent. So this is what we could call a generative process.
And, uh, uh, uh, generative model is as follows. So this, this is some kind of where an agent
has, uh, maybe some kind of simplified version of what's happening in the world. It is representing in
its brain. And, um, from that world, from that, some kind of simplified world state, it can predict,
okay, what kind of observations can I expect, um, from this state. And this is some kind of a very similar
model. And in both cases, we can also actually include actions. So if it acts on the world,
it will influence its next world, some kind of simplified world state. And it's the same also here,
it will also influence the real world state. So this is called a generative model.
And, um,
this generative model, it uses it to, um, to choose its next action. But then the second question is,
how does it learn, uh, uh, uh, how does it learn this model?
So, um, this papers, basically, we can basically divide it into two parts. One part is called inference
and the other part. So this is, we could say also action selection

and another part is called learning.

The generative model.

So, um, this presentation will also be, um, somehow divided into these two parts. So I'll first

go further into this first question. How does the agent select its actions?

And then also, uh, the second later on, I will go on to in the second part, how this is learning the generative model.

And, uh, please, if anything is unclear, please feel free to ask questions.

If you're now in the live chat, or if you're watching this later,

you're also always welcome to send me an email or get in touch in another way.

Um, if you have any questions, I'm really happy. So I wrote this paper for people in my situation three years ago, who really want to know the exact mathematical details of how active inference is formulated and, uh, want some kind of an accessible, easy overview. And, um, if you have any suggestions for improvement, I'm also very open to hear them. So, uh, yeah, so that's the basic overview.

And what I have to say as well is that this is all, as you saw also from the symbols and stuff,

this is all happening in a discrete space, uh, discrete observation space, action space, and discrete time.

So there's a whole other literature on the continuous side of active inference. I'm not going into that right now. So this, that's also not what this paper is about. It's only about the discrete time, um, formulation.

Um, yeah, I'm not sure if we already should stop here for some questions or that I just, uh, continue going deeper into the, yep. Continue. Thank you. Okay. Yeah. So maybe, so one point, maybe I should spend a little bit more time on it is this idea of the generative model. So, um, maybe I write a new one here. So what I think is important to emphasize that this, these states S are really, uh, inside the brain.

So we could say, uh, this is really in the brain of the agent

and, um, um, from these states it's predicting observations. And usually this is some kind of simplified, uh, state of the world, and it needs to learn these states without anyone telling it. Okay. So maybe in machine learning or something you're familiar with supervised learning where you get an observation and, uh, uh, then it might learn which state do I have to associate to it. That's not the case here. So the agent really needs to learn itself. Okay. What is a good, uh, state to somehow encode this observation in so that I can do a useful planning. So this is, you could call unsupervised learning and there is no way of saying, okay, the real world, for example, is generating observations and therefore this state should be somehow related to that, uh, the real world state.

Yeah. And to also, uh, maybe, uh, make a bit more clear how it can be simplified. So for example, we could have an agent, so we could have a world in color, for example, that, uh, like there's all different colors in this world, but for example, an, uh, agent that can only see, uh, light intensity, but no color will have also a simplified, um, simplified, uh, state of the world, so to say, so that this will only be black and we don't have gray, I see, but this is white and this is gray.

Okay. So that's just to, um, just to emphasize this point. Okay. This, uh, generative model is really something that's inside the brain of the agent and it can use it to, for example, plan ahead.

Um, yeah, then I'll go back to the paper.

So, um, I will start with some notation. So, um, yeah, so what we are considering is an agent, which, um, which is going through capital T time steps. So it starts at time one

and, um, has also some kind of finite, uh, time T and it's performing actions the whole time and, um, receiving observations

so that we can also see here. So we have observations and I'm using

Tau, the Greek letter Tau for any time point and T for the current time point. So you'll see that

appearing more often this distinction in the, in the paper, and we can use this subscript, uh, with the column to, uh, have like a sequence of observations, for example,

and then very important. There's this word policy.

and actually if you come from a reinforcement learning background this has a little bit different

meaning but here in this context a policy is a sequence of actions and more specifically a

sequence of future actions so that is sometimes i find in the other literature not so clear

but here i want to make that very specific so we have a policy pi t and that is actions small t so

there is also small t so action small t up to big t so a sequence of actions and pi will be then the

whole from time point one and so usually i write a for actions that are already before performed in

the past and pi for the future actions that it somehow needs to do planning on um yeah and we have

this generative model which we write like this and so it's a probability distribution over observations

and states given actions and this theta is the parameter of the of the generative model

and in this first part we leave this theta we leave it out in the notation because we are assuming okay

it has learned this generative model and we're just interested in how it's using it and then in the

second part we will actually look at we will include it again to see how it's learning this

um yeah and then also a big part of active inference is of course that it needs to infer

in what kind of state it is so what kind so it has an observation and it has this generative model

and then it somehow needs to infer okay what kind of state could i be in

and um we use this that's also in general in the literature we use this letter q so for example

an agent receives this observation o1 and then wants to find out okay or express its

belief in which state it is and that we write for example like this

so this is the belief so this is a probability distribution over states

and it has as information this first observation and in general that's always

known to be known to

uh i can also show that here in the paper so so this is the the actual generative model here

and here we have this belief over states and what we are doing to make the notation a little bit
more concise is to write qt for example qt of st let's say then this t is implying all the information
that it has already from the past so this is we can also write it qst given all the observations
it has made and all the actions that it has performed so this t is just saying okay we include
in this belief all the information all the observations all the actions to form this belief so to say
um yeah so i think i have discussed now the most important things from the
the notation then we can i can have a break here also if there's any questions or otherwise i just
continue
sounds good continue thank you okay yeah so then we immediately so that was also one of my goals of
this paper is to immediately get down to this main question okay which action is the agent going to
perform and we immediately arrive there already so um
the action selection procedure can be described as follows so we have here a probability distribution
of the action over policies so policies were sequences of future actions
and we say according to active inference and agent selects its next action
by sampling by sampling a policy pi t from this distribution and selecting the action a t corresponding to that policy
so i can write it here again so we have this distribution for now we can
uh uh ignore a little bit um how exactly what exactly this um how this distribution comes about i will get to that in a minute
but we can just say okay there is some kind of distribution over policies over future action sequences
and we sample some kind of pi t star we sample it from this distribution so we have this pi t star
which is a sequence of a t star up to a t star and then we are performing
this action
this action 80 star and yeah you could say forget about the rest so this is all right i'll do it like
this this is somehow uh
we are not interested anymore so we are only performing this action 80 star and then we go to the next time
step get the next observation and then we'll do this whole process again so this for example this uh
action this action a t plus one is somehow discarded so for the next time step we'll do the process again
and then take the first action from that sampled policy
and that in a very basic way is already some kind of answer to this first question okay how is it selecting
its next action and what we see is so it has received these observations and perform these actions
and they also flow into this this probability distribution is dependent on this
um on these observations and actions and then the next question is okay how does it compute
this probability distribution
so um
that would be the second question to answer if we really want to get down to how is it selecting its actions
the first one uh the first one is the first one is the first one is the first one is the first one
which is the expected free energy and i have written one uh way of writing it written it here

one question there jesse um yeah why is the policy conditioned on actions from the past

uh when i don't see the a a1 through t minus one explicitly entering into the unpacking and policy inference

is based upon the observation coming in and how that affects the belief inference but how does or why do you

have a conditioning on a as well so the function g is dependent on a

so in order to compute this function g we need to know which actions have we done in the past

do we though yeah because it's but it's a bit hidden in the notation here so

we use these past observations to perform inference uh about for example here our belief in future

observations so in this t here in the q it's it's captured okay this is a belief about future observations

but with the information from observations from the past and actions from the past

it's using those to form the beliefs about the future

or here for example the same for future states

or here so in order to make this distribution what state will i be in in the next time step

um it is using uh some kind of belief about what state am i in in the current time step which then again

depends on what actions and what observations we have in the past

okay so in in practice the b transition matrix embodies the expected consequences of different actions but

here it's just written in terms of being conditioned upon the whole past sequence of observations and

actions that like would have led to that transition matrix being learnt

yeah so here we are assuming that the whole um that whole b matrix is already learned so that we can just

take as given and we only use this o and this a for performing state inference so to say so we need to know

somehow in what state am i now and then you can go back and think okay what actions did i perform in the past

and then that gives you more information about what state you could be in now

now and i wrote this very explicitly because often in other literature this dependence is somehow completely

not made clear and then i find that sometimes a bit confusing so that's why i wrote it very explicitly

okay in order to compute this distribution we need this information so to say so that's why i wrote it in this way

does that answer your question i think so i just still i just still don't i know it's sort of a small

thing but but and i i see that that's it's that you've thought about it it's just i don't see any of the past

time steps in the right hand side so

why would the action have that you took 27 time steps ago

other than how it influences how you believe the action to work now

why would the specific memory of the action

calculation a given number of time steps ago bear into the calculation now

so for example if we take this distribution right which is

our belief in what state we will be in in the next time step

then it i think there's many situations where that we can think of that it's relevant what i did

to steps to time steps before so for example if two hours ago i decided to fly to germany

then that will still have influence on where i will be maybe the next hour

but like in a grid world example if the transition matrix is the up down left right the fact that you moved
a certain way 25 time steps ago doesn't need to specifically come into play
the next consequences of your actions so let's say we have a grid world uh which is just 1d let's say
1d grid world and you can go left or left or right
2d grid world and you can go right those are the two actions
um and let's say this agent knows i'm always starting in location zero
and um let's say it uh or let's say differently here let's say here zero that's a bit easier
and then um every action you did in the past will have an influence on in what location
you're in now right if i for example we are now at t equals three then your location will be different
if a1 is left compared to if a1 is right
but is it conditioned upon your current location and the consequences of action from your current
location only or is it specifically conditioned upon an action that you took in a past moment
other than how that action resulted in you getting to where you are now
uh maybe i sorry that took maybe very long so these actions are really the actions we
performed in the past so not any random actions but really only the one that we performed in the past
is that your question or the past yeah i we we can continue i just i see the past actions matter for
changing where you are and if you're able and and you could do other learnings and so on but
the past actions themselves don't influence where you are or your beliefs about the consequences of
action which are all that come into play in the operational calculation of g
hmm which doesn't specifically invoke a past later okay okay we can maybe uh we can maybe postpone it
to later to see if we can yeah make it more clear so um yeah so maybe to get back to the
to the to the uh so we wanted to know okay what is this what will be the next action then we saw okay
there's this distribution uh that we sample policies from
and uh from there we take the first action and that will be our action and then we are now looking at
okay what is this function g that is used for uh for computing this um probability distribution
and that function g uh looks a little bit complicated so we can just now uh see what is all coming in
to compute this uh this probability distribution so we see um there is this uh this distributions q
which are our beliefs about future states and future observations so those are and we will discuss later
uh exactly how it can derive these uh distributions and they are all dependent on our policies or are the next
possible actions so for each different policy we'll also have different beliefs about what observations we
could uh get in the future so those are the main ingredients here and then we have this distribution
pc which is a preference distribution of the agent and this preference distribution i also wrote it here
it's completely distinct from the generative model p so it has nothing to do so p is a certain distribution
and then pc is a very different distribution so they have nothing to do with each other in that sense
but it is a an ingredient for computing this uh expected free energy
yeah maybe also nice to see so for example this uh distribute this distribution also in the paper of
uh lens dacosta and others uh we see it here in equation 10 so he also describes it here and actually
we don't see it it's just a black bar oh that's we do see the notes

right now it's just the tabs
and now yeah still just the tabs
um
it is weird yeah maybe reshare the screen yeah
did something change or yep okay now we see the paper
okay great sorry about that oh good and now you see the notes yep
and now you see the other paper there we go there we go yeah so this is the paper of lens dacosta
and others and he has here the same um distribution
over policies and then actually in thomas parr's book and others um
there the distribution is a bit more complicated they also have an e and an f term and i'm writing in the
paper uh i'm was planning not to go into that now but i'm writing in the paper
here also a note on why um they have those terms but then you can at least see okay
um this uh you can find this distribution in these i use those two as main sources
for writing this paper
um yeah so let's so maybe people who have already studied some active inference are more familiar with
what comes now but i thought just to briefly discuss uh this expected free energy and how you can think of it
so um we have the equation here and we see that there are two terms and this first term
um is a uh kl divergence between two distributions and this first distribution is distribution over our
beliefs in what state we will be in in the future given that we have some observations that which observations
are again are again appearing here and we compare this distribution so the beliefs about states given that we have
already the observations with this distribution where we don't have the observations
and um so what you could say a little bit is if these observations are very informative
then our belief about the state will be very different from when we don't have the observations
so then this difference will be large but if these observations are not so informative
then these two distributions will be pretty similar and then this difference will be small
small so the size of this difference says okay how informative will these future observations be
and these future observations depend again on the policy so we somehow take an average over
what future observations do we expect given that we have this as next actions and given that we already have
some observations and actions done in the past so this first term is called epistemic value of information or information gain
and then this second term we see this preference distribution here come up so this is the observations that the
the agent would really like to make and the distribution here is the observations that it thinks it will make if it is going to do these actions
and so um the higher this is basically the more likely this uh policy will give observations that it actually
wants to have so we want to have so we want this um and that we see um and that we see also
if we go back to this equation so we see the lower the expected free energy of a policy the more likely

it is that we will sample that policy and we also see that

the higher these two terms are because of this minus sign, the lower the expected free energy

will be. So we want both this utility and the information gain we wanted to be high

for a specific policy or those are policies that we are likely, the agent is likely going to sample.

So that is one way to think about the expected free energy. And then another equivalent way of writing

it is the following. So here, we actually don't have this minus sign. So these terms, we both want

them to be small, or the smaller they are, the more likely that policy is going to be sampled.

And this term here is an entropy. And it's basically looking at the entropy, or you could

also say uncertainty about the observation, if the agent knows the state it's in. So if it's in a

certain state, but even though it knows what state it's in, it is very hard for the agent to predict

what the observation will be, then this entropy will be high. And then so therefore also this ambiguity

will be high. And therefore, the agent prefers to be in states that have low ambiguity. So that's this first term.

And then the second term is again, KL divergence, but this time, between the belief that the agent has about his future observations, given a certain policy,

compared to the observations, it's actually wants to have so if this difference is big, then it believes it will get different observations, from the ones that it wants to have so that it says bad. So this term, which is called the expected complexity or risk, those

we both we wanted to be small. So that's a little bit of interpretation on this.

On this term, on this G that is used for calc for computing the distribution for sampling policies and therefore the actions.

So there is one more thing is that we can that is for the computation. Nice. So we can make some kind of approximations, we will come back to that later as well, some more detail. And then this G actually, we can write it as a sum of different of like single time step G

G taus. And that makes all the computations. And that makes all the computations a bit easier. And I know that there's much more to write or to say about this expected free energy and where it comes from. But I think that was not the aim really of this paper. So then I refer you to these other sources.

Yeah, so now we have discussed this distribution where the policies are sampled from we have discussed this G.

But then an important ingredient of this G of computing this G are these distributions q q t. So this belief about future or about states and about observations.

And also in the actual, and also in the actual, often we are actually interested in these. So not over sequence of states, but just over one specific time point tau.

And I was actually thinking for this podcast not to go into this too much, because I think it's relatively straightforward. So at least what I write here.

Because I think the philosophy of this whole approach is that we want to use some kind of Bayesian

Bayesian approach as much as possible. And what I have done here is just describe exact Bayesian inference, which would be somehow the simplest way of forming beliefs about states and observations.

And then actually in more advanced literature, so to say, you could say that is some kind of an approximation of this exact Bayesian inference.

So I would say you could read through this, I can maybe show one small example. So we have this generative model over observations.

states

model factorizes. So maybe I can just use what I already wrote down. So we see that we can really somehow decompose it into very simple elements where we have a prior over the first observation, this one. We have a distribution that is telling us, okay, if we are in a certain state and we perform a certain action, what state will be the next one? And we have a distribution that is saying, okay, if we are in a certain state, what observation do we expect to get? So those are the three ingredients which are also often referred to as the... Maybe I write on this also. So this is called the... Usually called the B matrix. This is called the A matrix. And this, I think, usually a D vector or something is prior. So those are the three ingredients for this generative model. And then, for example, so how you would perform exact patient inferences. For example, we say, okay, we risk... So we first can express our belief in what state we are in, we can just use, okay, that's the prior, which we assumed we already had. And then we then get an observation. So that's why this one is referring to. And that we want, in this case, to be the exact posterior S1 given O1, which is proportional to

O1 given S1. And our... So this is this A matrix.

So this is a very simple example of belief updating, so to say, which I describe in more detail here in the paper.

And I think if you're just starting out with active inference, I think it's quite nice to just think about

this whole state inference in this way and think of these other approaches like minimizing some kind of variation of free energy as some kind of approximations to just doing this exact patient inference.

So this is a very simple example of belief updating, so to say, which I describe in more detail here in the paper.

So this is kind of this is kind of this kind of ail

how it's choosing which next action it's taking. And so I'm skipping now over this learning part, we will get into that a little bit. And then here in section four, there's a completely worked out example. So I really recommend so you've might also seen this in, for example, Thomas Parr's book. I've tried to make it very explicit here and very precise that you can really follow exactly what is happening. And I invite you to just go through this example.

So this up till here is just some kind of definition of the model. And then here we say, okay, how it's performing the inferences how it's making these cues. And then here we see we compute the expected free energy for all different policies. So this is the first action. This is the second action. So we walk somehow through a an example episode of this mouse in this T mace.

which is acting according to active inference. And then second time step is updating its beliefs.

And then again, computing expected free energy and then again, selecting sampling a policy and its action. So yeah, so I invite you to go through that as well. And what I found hard, so I looked at this. There is also so in order to compute make these computations, I thought, okay, I really want to make the computation. So I looked into this library by MDP and also a little bit at the other one,

SPM, I think, which is written in MATLAB. But what I found hard was that in both code bases,
the actual computations, the actual computations, of what's really going on are very deep down in the code,
so to say so are not so accessible. And what I did was actually to, to write my own code to do the up to do
the these calculations, which is also available on GitHub. So here on the first page is the GitHub link.
And I can and what I think is relatively nice from this code is that it's very so you see the code, right?
Yep. Yep. Yep. So that it's relatively easy to read. So for example, there's a function sample action,
which is exactly and I have also tried to refer to all the equations in the paper. So for example, here,
this is the first equation we've been talking about this distribution over policies. And here you see
exactly how it's computed. So we have to compute for every policy to expected free energy. And in order
to have to expect it free energy, we need these beliefs over states. And then we have to take this
softmax function. Well, I did not actually say anything about so this, this sigma here, that's maybe
good to just clarify, this is the softmax function, which makes just these values. So we get for every
pi, we get a different value, makes these values into a probability distribution.

So for every pi, we have a different expected free energy, we make it into a probability distribution.

And then we sample from it. And then we take the first element basically of the policy, which is our
action. And then we return the action. So I think this code, if you're just interested, you can play
around with it, you can take some parts of it, you can do it, whatever you want. But if maybe if the
maths are not so clear for you, but it's more clear and code, then you can also have a look at this. So
it's has the same kind of sequence as the paper. So we sample an action house to expected energy
computed, we need these two things, info gain and utility. And here we have a part about how it's
updating its beliefs. And that's exactly the same as what we what is described here in the state inference. So this is a

little bit different, for example, from the pi mbp and, and SPM where they use more advanced methods for doing this state inference. But for easy examples, I think this is really more than enough, just to get a very precise idea of what is really happening.

And then there's also functionality for learning, which we will go into now.

So I think that's everything I wanted to say, at least from my side about the inference.

So maybe if there's any questions, I'm very happy to go into those now.

Okay, I could maybe maybe finish covering all the sections of the paper. And then there's a few questions in the live chat and people can add some more.

Okay, yeah. So for the second part learning.

I was thinking to actually go a little bit more or

to make it a little bit more difficult, so to say, and really go into depth of where the equations come from.

Because for a long time, that was actually also unclear to me and actually quite late in the process, it became exactly clear of how

how, where the equations for learning come from.

And I think that's also a quite a relevant or significant contribution of this paper of this paper to present that.

And I thought it might be nice for the people who are interested also in where they come from,

to go into a little bit more depth now.

So I would actually say that we go then to the appendix.

So there's here appendix B.

And we actually first look at a little bit simpler setting to somehow get familiar with the basics.

And then we go back to this setting of an agent operating in a world.

Yeah, maybe I introduce it a little bit here in the notes.

So I'm slightly changing notation now to avoid confusion, but maybe it also makes it a bit more complex. I don't know.

So we have now X and Z.

And you could basically think of X as an observation.

And Z as a

latent state, I call it so.

And maybe I just I think in most cases, you can also think.

About this as the O.

And this about the S, so to say of the of the model.

But that doesn't always hold. So that's why I chose to call it X and Z.

And we first for now forget completely about this.

About his set.

And we imagine some kind of very simplistic setting, where we want to where we are receiving observations X.

And we want to somehow make a model of these received observations.

So we want to make some kind of probability distribution X.

Checking which notation exactly.

I'm using.

I'm using.

Yeah.

So there's some different observations we can make.

One up to N.

And we want to model.

The observations we make with the probability distribution.

Which is dependent on theta.

And then very specifically.

So it is just the probability that we see an observation.

I given theta is theta I.

No.

So theta is a vector from theta one up to theta N.

And the learning.

And then you could say learning is finding this theta.

And what the active inference literature is suggesting is to think of this theta also.

So that is a very basic and kind of approach that we think of this theta also as something

that we can have a belief about.

So we.

On its.

So we can have also a probability distribution over the theta.

Which is somehow expressing.

Okay.

I believe my belief about the theta.

That theta I is 0.1.

For example.

Is.

Is this number.

That is known to be

And so alpha is also a vector from alpha one to alpha n.

So this distribution, and by the way, there's another paper from Ryan Smith, I think it's also in the references, which also describes this gives a little bit more intuition about this whole process, which I think is really nice.

I can also recommend looking at that.

Yeah, but so this is basically describing our beliefs about our parameters.

And then just so we could say the prior belief.

And then we can, for example, we make a specific observation x star.

And then we can ask ourselves.

How is this?

Believe go to change if we incorporate this new information that we made a certain observation.

And that I'm not going to write down, but we can look at here.

Yeah.

Yeah.

So what I just described was this.

So we have this distribution parameterized and this belief about our data.

And then we can look at the different alpha.

And then we are interested in this posterior belief, so to say, after having this made this observation x star.

And what we can, if you work that out, you see that that's again a Dirichlet distribution, but then with a different alpha.

So we call the updated alpha, we call it alpha prime and this.

This thing here is the indicator function, which is basically, which is one when x star is equal to x i and zero otherwise.

So what is happening is that we increase the alpha i by one which corresponds exactly to the observation to the eye of the observation and the rest of the alpha stays the same.

And if we look again at this definition of the Dirichlet distribution, so we see the higher the alpha, the more weight that theta i is going to get.

In the probability distribution.

So this is a very, I would say, elegant way of updating your beliefs about the theta.

When you get new observations.

And this is a very, I would say, and this is the general idea of how the theta is also learned in the setting that we are looking at it.

This, uh, pom DP, this partially observed mark of decision process setting where we have states.

So hidden states, uh, observations and actions.

Um, yeah.

But so what we were looking at here was just a setting where we had just observations, but we didn't have any latent states.

And now the question is what happens when we do include latent states.

So we can do a similar derivation.

So we now have two distributions, uh, depending parameterized by two different thetas.

So we have prior, which is a vector over the latent states.

And then we have some kind of matrix, which is saying, okay, given we have a certain latent state, what's the probability of a certain observation.

Um, and, um, then we can, again, ask ourselves this question, um, what happens with our belief over theta?

If we get an extra observation.

And what is actually, uh, the case here is that we get some kind of very complicated distribution, which is not, uh, a Dirichlet distribution.

Um, so this exact method is not so useful for updating.

So this is more like the setting.

We, uh, we are looking at, right, where we have observations and latent states.

Um, so we have to do something slightly different.

And, uh, that is, uh, that we do by the mean field approximation.

And, um, so what we do instead is we, um, uh, let's see.

Let's see what, how I can best continue.

So maybe.

I go very briefly briefly to appendix a.

Appendix a.

So in appendix a, I, uh, describe some kind of very general.

So I think most of you have already seen this, but just for the completeness, uh, a variation of free energy minimization approach.

So, um, um, it's described here.

So we have some kind of, uh, joint distribution and we want to, uh, uh, computer posterior, which is also the setting we had before.

Um, but often this, uh, posterior is hard to compute directly.

So what we do then is we instead, uh, choose a family of distributions, Q.

And, um, find.

So this Q tilde are all element of this Q.

And we try to find, uh, that Q tilde that is minimizing this expression.

And that Q, Q, we call Q X because it's actually dependent on this X on the observation that we made.

And this is some kind of very general approach, which is called the variational approach.

And that is somehow used to approximate a posterior that we can't, uh, compute directly.

So that is described in more detail here in this appendix and also how it's plays a role in active inference.

Um, I think for time reasons, I won't go into that much deeper now.

Um, but just.

To describe this method at least.

So we use that same method also here.

So right now we are dealing with latent states.

So these sets that we don't know, and we are dealing with the parameters that we don't know.

So we somehow, uh, but what we.

Do know is the observation and the hyper parameters.

And we want to find the distribution over the latents and the parameters.

And then we use the same, uh, variation of free energy minimization approach.

Where we are trying to find the queue.

That is minimizing this distribution.

And then what we do, what I was also saying before, often we restrict the queues that we can choose from.

Which is, uh, and in this case, we, uh, are only looking at queues.

Which can be written as a product of these individual distributions.

So that's some kind of an approximation that we are making here.

Which is called, uh, the mean field approximation.

And, um, if we do that.

Then, um, this, uh, this equation simplifies a bit.

And what we also do.

So this is, uh, very important insight.

Is that it's quite hard to minimize.

Both of these queues at the same time.

Because they are both influencing each other.

And there is a specific algorithm called the CAFI algorithm coordinate ascent variational inference algorithm.

Which is saying, okay, instead of trying to find both at the same time.

We start by fixing one.

And then optimizing the other.

And then use that.

Uh, so for example, we start by fixing this cute theta.

And then find Q set.

And then use this newly found Q set to find an Q theta.

And then you can do that iteratively and hopefully get, uh, to the optimizer.

And that is also exactly the approach that is used for learning in the active inference.

So very specifically, we first have some kind of prior belief.

Belief about our generative model, about the parameters of the generative model.

So we fix those, uh, that belief, so to say, or those parameters.

And with those parameters, with the generative model, we can, uh, do inference.

Like, uh, we looked at before, like this base and, uh, base and inference.

And that's also what's done here.

So we, uh, get some kind of Q set.

So that is, so this is a different way of describing inference in terms of variation of free energy.

But what's basically what's done here is we are just trying to infer what kind of latent state are we.

And we have fixed the parameters Q theta.

So, and then we have this, uh, Q set.

So this belief about the state we are in, and then we are somehow where we want to be.

We want to, to find this distribution, uh, Q theta, and then we use these, this fixed Q set to then optimize this distribution, which is only dependent somehow on Q theta.

And the Q set is fixed.

And if you would really want to find the minimizer, you would continuously be doing this alternating these two steps.

Um, and, um, what is very important to note here is that, uh, the active inference literature is saying our update is only doing the first step.

So we only do the first step.

So we only do the first step.

We do one time state inference and one time, uh, we optimize, uh, this parameter belief on the parameter given this state inference.

And then we stop.

So that is, so it's not exactly the same as the CAFI algorithm, but you could say it's only the first step of the CAFI algorithm.

Um, and, um, so here I'm describing what does that minimizer then look like.

So, um, and that is given by this, um, basically the exponent of the term that's here.

So you can also see that if we take the exponent and we take the log, we just get what's there.

And then it's minus what's there.

So that is the minimizer, so to say, and I'm using these terms C with the backslash to say, okay, these are all terms independent of the argument.

So we don't have to worry about them or the minimization.

So here we have a very specific form of the updated belief in our parameter.

So, um, and, um, then we can use the fact that we are actually working with this, this, uh, Dirichlet priors and this categorical model.

So, um, and then we can, so that P here that was defined here.

So in terms of these Thetas.

And in terms of these Dirichlet's.

And what you see then is that this, uh, updated queue after the first step in this algorithm, uh, looks like this.

And this is exactly again, a Dirichlet distribution.

So that's a really nice result.

So I didn't make any assumption that it has to be a Dirichlet distribution.

So we only said, okay, we do this mean field approximation,

but it comes out that it is, if you only do the mean field approximation,

that is first step of the CAFI algorithm is again, a Dirichlet distribution.

So we can, um, formulate the update again in a very elegant way like this.

So we just add something to these two updated alphas.

And what we add exactly is the following.

So for example, this, uh, alpha D is related to the Theta D, which is, um, um,

which is to believe about the prior belief in the latent state.

And we add, um,

we add the belief after, uh, doing the inference that we are in that state.

So this is in some sense, a very similar update to this update here,

where we are just adding one for the observation that we did,

but now we are not adding one for the observation,

but we adding only the belief that we are in that state to,

to the, uh, to the original alpha.

And something similar here where we add the belief in the transition more or less.

So, um, yeah, I can imagine that this was a little bit fast and maybe a little bit abstract

because there were not, not so many examples,

but I felt this for the people who really want to know, okay,

where do these equations come from?

I think this is quite a, quite a, uh, relevant contribution.

And, um, so what I was talking about now was only this X's and Z's.

The notes are still there, right?

Yeah.

Great.

Um,

And now we want to translate this to this setting where we are actually, uh,

having this S and the A's and the O's.

So then the equations get even a little bit more complicated.

So I thought it was good to have this build up to first see, okay,

what is the general principle?

This mean field approximation and the Dirichlet coming out,

and then you can apply exactly the same procedure to this more complicated setting

where we have this full generative model.

And then you get this derivation here, which I don't, uh, I'm not going into now,

but there we see again at the end that we have a Dirichlet distribution.

And here you see, okay, alpha, we have to add this to the alpha.

We have to add this to the alpha.

A we have to add this to the alpha B.

And then we get, so this is the, the main part of the paper.

Uh, we have these, uh, updates here for the learning.

Um, so, um, yeah, so for the people interested in, uh, knowing where do these learning rules come from,

um, there is this appendix B, which I went through now a little bit quickly and maybe a little bit high level, but, um,

yeah, you can look at that in more detail or ask me questions about it if you, uh, are interested in this.

And maybe to just link it to the other literature.

So here there's, this is the book of Thomas Parders is appendix B two, three learning.

And here he is coming to the same, uh, equations, but using a bit different method that I found hard to follow.

So, uh, yeah, but these, these update rules, they are also here and, uh, the same also in this paper.

Let's see if the reference.

Yeah.

So here we also have these, uh, updates, but also using a different method than the one I'm using.

And actually this method that I am presenting is referred to me by Connor Heinz.

So thanks very much for, uh, pointing me to this derivation, because I think this for me, at least is the most clear one I have seen so far.

Um, yeah, so, so far about, uh, learning.

So there's one, I would be able to, if there are time allows, could go briefly over the appendix a, but maybe we also first take some time to, uh, for questions about this learning part.

And I'm also fine with not discussing, uh, this appendix a.

Awesome.

Okay.

I will, uh, ask some questions that were submitted by Aaron.

Aaron.

And then if anyone else has any questions, they can go for it.

Okay.

First question.

Do you intend the accompanying repository to be a static accompaniment to the paper?

Or do you see it evolving?

For example, adding habitual policy selection to the agent via e tensor.

Um, I would say from my site, uh, I'm not really planning on, uh, developing it further, but, uh, if people want to, I am very happy.

For example, if they fork the repository, copy it in any way would be nice if you reference me, if you still use it, but, um, yeah, so from my side, I don't see any further developments.

Uh, but I would be very happy if other people want to actively start using it.

So please get in touch if you want to.

Cool.

Okay.

Would it be possible to further document the flattening class at first with.

At first with comments, but ideally with unit tests.

Yeah.

So that is indeed.

So, uh, maybe a good point to make about the, the code.

So, um, in PI MDP, uh, there is some, I have that also here in appendix C, I think.

Um, some, yeah, there is some extra, um,

um, kind of structure being put on, for example, these, uh, uh, these conditional probability distribution.

So for example, we say.

Um, the location in the next time step is independent of the reward condition in the previous time step.

And that is for the computations.

It's very nice because the computations get a bit, uh, easier.

Um, but, um, the code gets a lot more complicated and you don't see, especially because for example, here, um, if we look at all these, uh, for example, get believe next states.

Yeah, I think that's a good example update, believe current state, something like this.

Um, for example, this one, um, this is very simple and elegant, I would say.

And that is because I am ignoring that, so to say in the code that, that there are excessities in dependencies.

And I did that, uh, uh, on purpose to keep the code as simple as possible.

But what is then a bit harder is that, so, um, so what, uh, uh, PI MDP is using.

Is, um, having some kind of state, which is somehow, uh, multi dimensional.

And then also, for example, uh, the, a matrix, let's say, which, um, is also has is, I think, some kind of, uh, list.

So in the observation state, so there's observation modality state factor.

So these are both have can have multiple, uh, elements.

So this is a list of matrices and these matrices.

So for each modalities, let's say, each modality M.

And, um, this is then again, um, some kind of tensor where we have here.

The, uh, observations for that specific modality, but then we still have.

Here, uh, also for every state factor, uh, a different, um, dimension, so to say.

And in my case, what I am doing is just saying, okay, we just say that we, the state space is, let's say one dimensional.

And we just have, uh, uh, S one, one.

One.

S, uh, one, two, let's say there's two state factors.

And I'm just putting them all into a long list.

So to say, so that's what I call flattening.

And then this A matrix is just one matrix where there's all the, the flattened states here.

Let's say S flat.

And here, O flat.

And that makes the, the code very easy, but there, I do need something to translate between the PI MDP states, which are somehow have these different state factors and the flat states that I am using.

And that is indeed, I agree is not, uh, uh, uh, the best documented code.

So I, I wrote that and then I tucked it away in this flattening, uh, class and was hoping that everyone would believe me that, uh, it works well.

But we could talk about it because I do think it's a good point that it would be nice if it has like, uh, if you don't just have to, uh, believe me that it works, but that you can actually check it yourself.

So maybe we can, you can send me an email and we can discuss how to best, uh, to best documented.

Yeah.

Yeah.

Lots of, uh, little challenges with matrix format and small things that, uh, yeah, definitely.

Okay.

I'll ask Ernst last, last, uh, question here.

And this is kind of a good, uh, final discussion piece, which is what do you see as the next key area to tackle in terms of increasing the cliff?

In terms of increasing the clarity of how active inference works.

Hmm.

Yeah.

That's an interesting question.

So I felt that with this paper, most of the questions I had were answered.

Um, I'm now looking a bit, uh, together with, uh, Lance D'Acosta also at how well, for example, these learning rules in practice are actually performing.

Um, but that is maybe different from, uh, uh, from, so I feel at least, uh, in my opinion, in the paper, it's clear how they are formulated.

So I would say, yeah.

Um, I would be open for suggestions, but I, so I have answered with this paper.

Most of the questions, at least I had.

About how active inference works.

And then of course about this very specific domain of the discrete time, uh, domain.

Yeah.

How, what, what did you, uh, think or have any conversations about how this relates to the continuous and the path-based versions?

So I have actually only spent time with the discrete setting.

So, um, I was, I think in the beginning, very optimistic.

I thought, okay, first understand the discrete and then the continuous, but then I first spent time on understanding it.

And then I thought, okay, let me write this paper on it, which also took quite some time.

So I actually have no experience with the continuous case.

I'm not sure.

Lance is not here, right?

Or, um, no, I don't, I could ask him, but, uh, he is more familiar with, uh, so I think that's more a question for him.

Maybe he said some way able to answer that at some point.

Cool.

What are your next steps or how are you going to move forward?

Having satisfyingly reduced your uncertainty about discrete states?

Yeah.

So I am very interested in general in this, uh, we could maybe call it a B learning, uh, domain.

So how is an agent actually learning these, uh, matrices?

So, uh, uh, together with Carlotta, with my colleague, we were also applying this framework to study, uh, consciousness or are still busy with that.

And, uh, paper about it, hopefully coming out soon where we, uh, use an active inference agent, and then, uh, look at how it's, uh, integrated information is, uh, developing.

And, um, but we saw actually in the environment that we were using, that it was very hard for the agent to learn the A and B matrices or learn an A and B matrices that allows it to, uh, operate well in the environment.

So for me, that's the next big question is how, what is actually, are these learning rules from active inference, the right ones, or are there other rules that are maybe more effective and maybe more probable.

So that is for me, a big question that I completely don't have an answer to right now, but this really, I find really fascinating.

Awesome.

Any other comments you'd like to make?

Um, so I think it's good to finish it here, but just to, to say that it's, there is, there's also this appendix A.

So maybe a bit too weirdly, but the, uh, uh, I think all of free energy and active inference usually is formulated as a minimization of this variation of free energy.

So, but starting the story from there, I find, uh, but starting the story from there, I find then it gets very confusing, uh, most of the time, but I do write here in appendix A, how everything that's written in this paper can also be related.

So, uh, to, uh, to, uh, to, uh, to this bigger idea of minimizing and, uh, variational free energy function.

So you can just read through it here, perception, learning and action selection, how they are all related to minimizing this function.

Just so you know that it's there.

Awesome.

Awesome.

Well, again, thank you for the importance and pedagogical work.

I look forward to many in this genre.

Thank you so much for the opportunity.

And, uh, again, if anyone is, has, has any questions or wants to reach out, please feel free to do so.

Awesome.

Thank you.

See you.

Thank you.

Thank you.

Thank you.

Thank you.